An Introduction to Differential Equations


Leibniz


Newton

An <u>ordinary differential equation</u> is an equation that involves a function and its derivatives.

<u>Ex</u>. Newton's Second Law $F = ma$

Let $x(t)$ be the position at time t:

$$F(t, x, x') = m \cdot \frac{d^2x}{dt^2}$$

Our Approach to Differential Equations:

Physical System

Construct a DE → Solve the DE

[instantaneous info → long-term info]

<u>Ex</u> (Radioactive Decay). Let $x(t)$ be the amount of radioactive substance at time t.

$$\frac{dx}{dt} \propto x \Rightarrow \frac{dx}{dt} = -kx \quad [k > 0]$$

$\frac{dx}{dt}$: $\frac{[\text{mass}]}{[\text{time}]}$; x: [mass]; k: $[\text{time}]^{-1}$

<u>Goal</u>: Solve – find $x(t)$ that satisfies the equation!

There is an "obvious" solution: take $x(t) = 0$.

Check: <u>LHS</u> $\frac{d}{dt}x = \frac{d}{dt}0 = 0$ ✓

<u>RHS</u> $-kx = -k \cdot 0 = 0$ ✓

If $x(t) > 0$ ...

$$\frac{1}{x}\frac{dx}{dt} = -k$$

Chain Rule: $\frac{d}{dt}\ln x = \frac{1}{x} \cdot \frac{dx}{dt}$

$$\Rightarrow \int \frac{d}{dt}\ln x \, dt = \int -k \, dt$$

$$\Rightarrow \boxed{\ln x = -kt + C, \; C \in \mathbb{R}}$$

Exponentiating, we have

$$x(t) = Ae^{-kt}, \; A \in \mathbb{R}$$

We obtained a general solution to the eq., which involves an arbitrary constant.

Separation of Variables

Ex. Consider the differential eq.

$$\frac{dy}{dx} = e^x(1+y^2).$$

Separate the variables:

$$\frac{1}{1+y^2}\frac{dy}{dx} = e^x$$

Chain Rule: $\frac{d}{dx}\tan^{-1} y = \frac{1}{1+y^2}\frac{dy}{dx}$

$$\Rightarrow \int \frac{d}{dx}\tan^{-1} y \, dx = \int e^x \, dx$$

$$\Rightarrow \boxed{\tan^{-1} y = e^x + c, \quad c \in \mathbb{R}}$$

We get the same result by "separating" dy and dx:

$$\int \frac{1}{1+y^2}\, dy = \int e^x \, dx$$

Ex. Consider the initial value problem

$$\frac{dy}{dx} = \frac{xy^2}{x-1}, \quad y(2) = 1.$$

Separate the variables:

$$\int \frac{1}{y^2}\, dy = \int \frac{x}{x-1}\, dx$$

$$\boxed{-\frac{1}{y} = x + \ln|x-1| + c, \quad c \in \mathbb{R}}$$

We require that $y = 1$ when $x = 2$: find c!

$$-\frac{1}{1} = 2 + \ln(2-1) + c \Rightarrow c = -3$$

$$\boxed{-\frac{1}{y} = x + \ln|x-1| - 3}$$

We have a particular solution to the diff. eq. that satisfies the given condition.

Ex. Consider the differential eq.

$$\frac{dy}{dx} = \frac{1-2x-3y}{4x+6y+2}$$

⚠ This is NOT separable!

$$\frac{dy}{dx} = \frac{1-(2x+3y)}{2(2x+3y)+2}$$ Let $u = 2x+3y$, $\frac{du}{dx} = 2+3\frac{dy}{dx}$.

$$\Rightarrow \frac{du}{dx} = 2+3\left(\frac{1-u}{2u+2}\right)$$

$$\Rightarrow \frac{du}{dx} = \frac{1}{2}\cdot\frac{u+7}{u+1}$$ This is separable! ∴

$$\Rightarrow \int \frac{u+1}{u+7}\,du = \int \frac{1}{2}\,dx$$

$$\Rightarrow u - 6\ln|u+7| = \frac{1}{2}x + C$$

Since $u = 2x+3y$:

$$\boxed{(2x+3y) - 6\ln|2x+3y+7| = \frac{1}{2}x + C}$$

This solution requires that $2x+3y+7 \neq 0$.

⚠ Is $2x+3y+7=0$ a solution?

We check if $y = \frac{1}{3}(-7-2x)$ satisfies the eq:

LHS: $\frac{d}{dx}y = \frac{d}{dx}\left(\frac{-7-2x}{3}\right) = -\frac{2}{3}$ ✓

RHS: $\frac{1-2x-3y}{4x+6y+2} = \frac{1-(-7)}{2(-7)+2} = -\frac{2}{3}$ ✓

Yes! We have a singular solution

$$2x+3y+7=0,$$

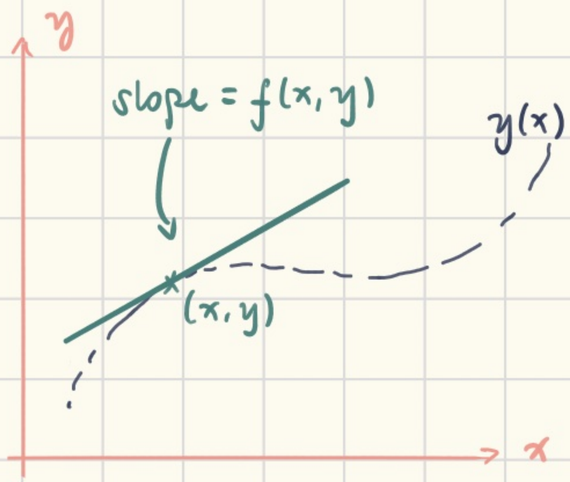
which is not captured by the general solution we found.

Geometry of Differential Equations

Consider the eq.

$$\frac{dy}{dx} = f(x,y).$$

Given a solution $y(x)$, $f(x,y)$ is the slope of the tangent line at (x,y)



We sketch a slope field to investigate the behavior of solutions.

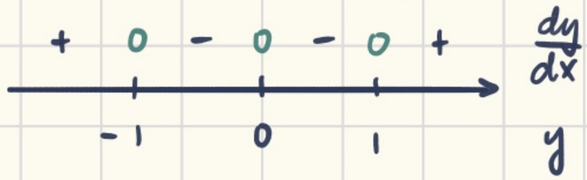
Ex. Consider the diff. eq.

$$\frac{dy}{dx} = y^4 - y^2 = y^2(y-1)(y+1).$$

Equilibria are constant solutions, which occur when $y'(x) = 0$:

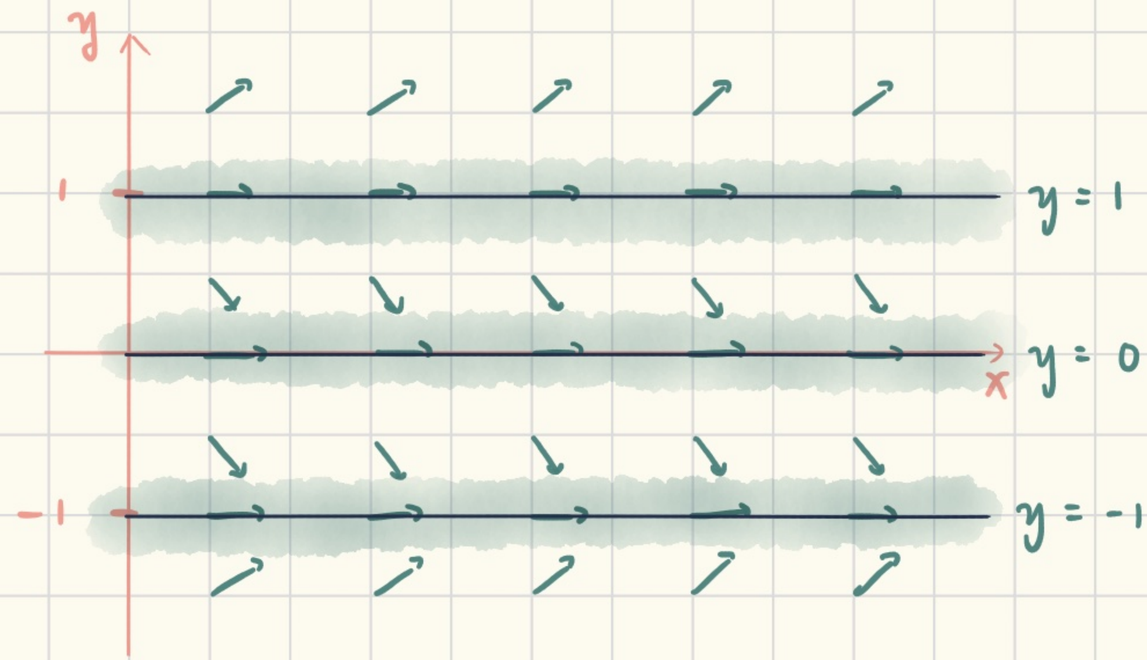
$$y(x) = 0, \quad y(x) = -1, \quad y(x) = 1.$$

We check the sign of y' elsewhere:



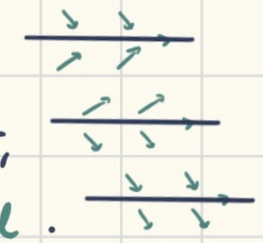
We sketch slopes:

- ↗ when $y' > 0$,
- ↘ when $y' < 0$,
- → when $y' = 0$.



We classify the equilibria according to their stability:

- $y = -1$ is stable;
- $y = 1$ is unstable;
- $y = 0$ is semistable.



Population Dynamics

Let $y(t)$ be the population of some species at time t, with $y(0) = y_0$.

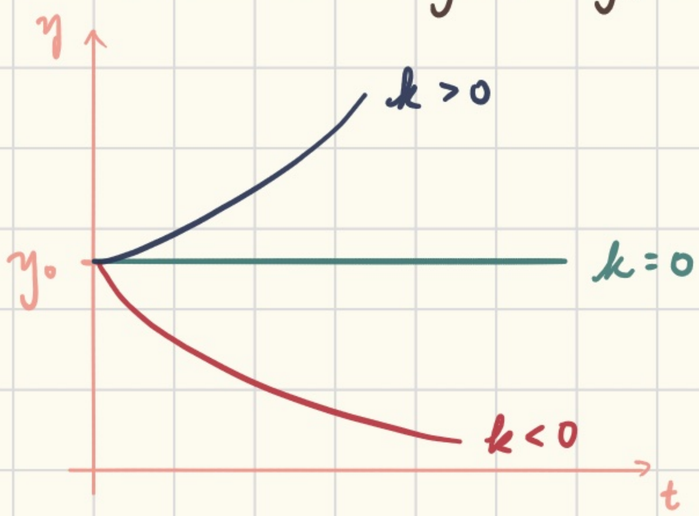
How do we model $y(t)$?



Malthus' Model

$$\frac{dy}{dt} \propto y \Rightarrow \frac{dy}{dt} = ky$$

Solution: $y(t) = y_0 e^{kt}$





Verhulst's Model

$$\frac{dy}{dt} = \boxed{k}y$$ → modify k to depend on y!

$$\Rightarrow \frac{dy}{dt} = k\left(1 - \frac{y}{\beta}\right)y \quad [k>0]$$

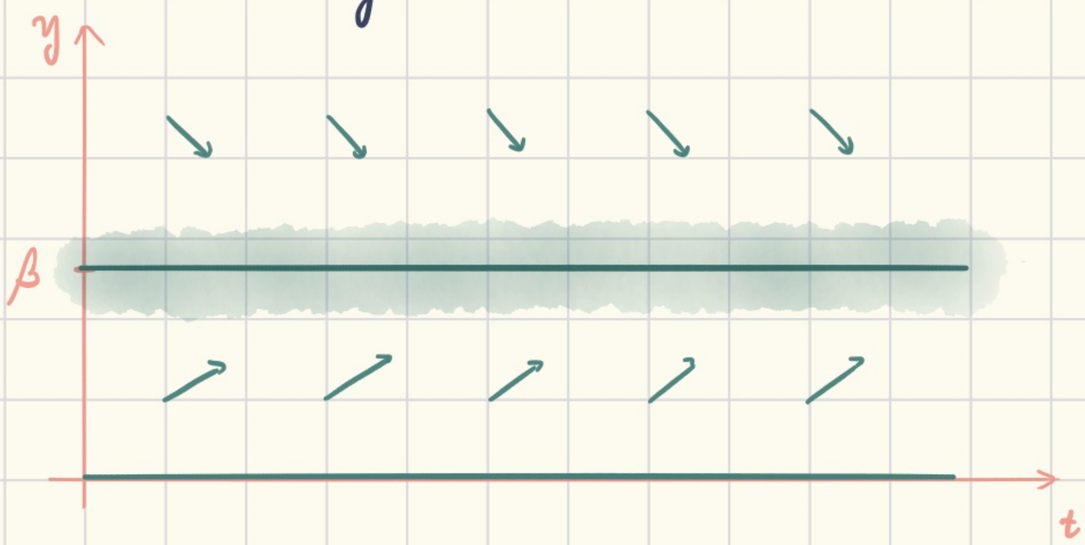
When $y \approx 0$, $y' \approx ky$;
when $y \approx \beta$, $y' \approx 0$.

We sketch a slope field!

Equilibria $[y'=0]$: $y=0$, $y=\beta$

Check: $\frac{dy}{dt}$





$y = \beta$ is a stable equilibrium!